

FREE GUIDE FROM BACKUP POWER PRO

The Homeowner's Guide to **Generator Connection** in SC

What to Know, What to Ask,
and What It Really Costs Before You Call Anyone

7-minute read, everything you need before getting a quote

A practical guide from a licensed SC electrician.
No filler. No sales pitch. Just what you actually need to know.

Backup Power Pro, A division of Key Electric

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Why Most Generators Don't Actually Help During an Outage

The problem isn't the generator. It's the connection.

If you own a portable generator and it's sitting in your garage right now, you're not alone, and you're not as outage-ready as you think. Most generators in Upstate SC get used about twice a year: once when the homeowner fires it up to test it, and once when the power actually goes out and they scramble to figure out what to plug into it.

The Real Problem

Without a proper inlet and interlock installed at your electrical panel, your generator can't power your home's essential circuits, the lights on the wall switches, the refrigerator that's been running for years, the furnace blower that needs a thermostat to kick on, the well pump or sump pump that are wired directly into the house.

Running extension cords from a generator outside to appliances inside is not a connection. It's a workaround, and a dangerous one.

Extension cords create real risks:

- ⊗ **Backfeed hazard:** Without an interlock, generator power can travel back into the utility lines, creating a lethal hazard for linemen working to restore power.
- ⊗ **Undersized cords:** Most household extension cords aren't rated for the load a generator puts out. Overheated cords are a house fire waiting to happen.
- ⊗ **Code violation:** This method violates NEC electrical code and is excluded from most homeowner insurance policies.
- ⊗ **CO risk:** Running the generator too close to any opening to the home can allow carbon monoxide inside.

What you actually can't power:

Even with a generator running at full output, extension cords can't power:

- Lights on wall switches
- Refrigerator and freezer (wired circuits)
- Well pump or sump pump
- Furnace blower motor
- HVAC systems
- Garage door opener
- Built-in appliances

THE BOTTOM LINE

Most homeowners who buy a generator spend \$800-\$2,000 on the machine and never make the \$1,200-\$1,500 connection that makes it actually work. This guide explains exactly what that connection is, what it costs, and what to look for before you hire anyone to do it.

The 3 Ways to Connect a Generator to Your Home

And what they actually cost, side by side.

OPTION	ALL-IN COST	WHAT IT POWERS	CODE LEGAL	TIME TO INSTALL	BEST FOR
Extension Cords	\$0-\$150	Only what's plugged directly into the cord	✗ No	None	Emergencies only, not a real solution
Generator Inlet + Interlock Kit SMART CHOICE	From \$1,197	Lights, fridge, well pump, sump pump, furnace, Wi-Fi, phone charging, your essential circuits	✓ Yes	1 day. Permitted. Inspected.	Homeowners who already own a portable generator
Whole-Home Standby	\$12,000-\$20,000+	Entire home, automatically, no manual start needed	✓ Yes	6-12 weeks	Medical equipment, home offices, or those who want seamless backup

Option 1: Extension Cords, How Most People Do It Now

Cost: \$0-\$150 in cords **Code compliant:** No **Insurance covered:** Usually not

It's not a connection, it only powers what's plugged directly into the cord. Your lights, furnace, well pump, and fridge stay dark. Backfeed risk, fire risk from undersized cords, and NEC code violation. Most homeowner insurance policies exclude generator damage from this method.

Option 2: Generator Inlet + Interlock Kit, What BPP Installs

Cost: from \$1,197, all-in **Includes:** Inlet, interlock, power cord, permit, inspection **Timeline:** One day

A permanent outdoor inlet is mounted to your home and wired to your breaker panel. A mechanical interlock prevents backfeed, legally required. During an outage, you plug in your generator, flip the interlock, and choose which circuits to power. 80% of the protection of a full standby system at a fraction of the cost.

Option 3: Whole-Home Standby Generator, The Premium Choice

Cost: \$12,000-\$20,000+ **Timeline:** 6-12 weeks **Fuel:** Natural gas or propane (auto-start)

Turns on automatically within 10 seconds of an outage. Powers the entire home without load management. Runs on natural gas or propane. The gold standard, but at a price point that doesn't make sense for most homeowners who already own a portable generator.

What Actually Gets Installed, In Plain Language

Three components. One-day install. Code compliant from day one.

1

The Generator Inlet (Outdoor Receptacle)

A heavy-duty outdoor outlet mounted on the side of your home, near where your generator will be parked. This is what you plug your power cord into when the power goes out. It's weather-rated, tamper-resistant, and sized to the amperage of your generator. Looks clean, about the size of a dryer outlet on the exterior wall. This is the connection point that makes everything else possible.

2

The Interlock Kit (Mechanical Breaker Safety)

Installed directly inside your electrical panel. A metal slide mechanism that physically prevents your main utility breaker and your generator breaker from being on at the same time. This is what makes the whole system code-legal. Without an interlock, or a transfer switch, a generator can backfeed electricity into the utility lines and create a lethal hazard for the line workers trying to restore your power. The interlock is required by NEC code. It's not optional, and any electrician who skips it is cutting a dangerous corner.

3

The Power Cord

A heavy-gauge cord (typically 25 or 50 feet) that connects your generator to the inlet. Your generator stays outside, at least 20 feet from any openings to the home. The power runs through the cord, into the inlet, through the panel wiring, and out to the circuits you select. The cord is sized to your generator's output.

How You Use It During an Outage (6 Steps)

1

Move your generator outside, at least 20 feet from doors, windows, and vents

2

Add fuel if needed, start the generator, let it warm up for 30 seconds

3

Plug the power cord from the generator into the outdoor inlet on your home

4

Go to your breaker panel, slide the interlock to "generator mode" (this disconnects utility power)

5

Turn on the circuit breakers for the loads you want to power, refrigerator, lights, well pump, furnace, etc.

6

Your home runs on generator power through the circuits you selected

WORTH KNOWING

The whole process takes about 5 minutes once it's installed. There's nothing complicated about it, it's a panel connection you don't have yet. Once you have it, outages become an inconvenience instead of an emergency.

What Should Be Included When You Hire a Licensed Electrician

A complete installation should cover all of this, before you write a check.

A Complete Installation Includes:

- ✓ The inlet hardware, weatherproof, correct amperage for your generator
- ✓ The interlock kit, sized for your specific panel model (not universal)
- ✓ All wiring between the inlet and the panel
- ✓ A panel safety assessment before and after the install
- ✓ The power cord, sized to your generator's rated output
- ✓ Permit application filed with the county building department
- ✓ County inspection, scheduled, passed, and documented
- ✓ A full functional test before the electrician leaves your driveway

What a Quote Should Never Hide:

- ✗ **"Permit and inspection billed separately"**, permits are legally required. They belong in the price from the start.
- ✗ **"Additional charges for older panels"**, a licensed electrician should assess your panel before quoting. Surprises on install day are a red flag.
- ✗ **"We can skip the permit to save you money"**, this is common. It saves the contractor paperwork and creates liability for you as the homeowner.
- ✗ **No warranty on the installation work**, hardware has manufacturer coverage. The install itself should be guaranteed by the contractor.

SOUTH CAROLINA LAW

In South Carolina, generator connections require a permit and county inspection. This is not optional, it's state law under the NEC Article 702 (Optional Standby Systems) as adopted by SC. If a contractor quotes you a lower price and skips the permit, you're the one who owns a non-compliant installation. Your insurance may not cover generator-related damage or fire claims on an unpermitted system.

A Note About How BPP Prices This

Backup Power Pro includes the inlet, interlock kit, power cord, all wiring, permit filing, county inspection, and a panel safety check, in one flat price. No line items added after the fact. No surprises on install day. The price we quote is the price you pay.

Permits, Inspections, and Why They Actually Protect You

This is the part most contractors don't explain. Here's the full picture.

Why a Permit Is Required in SC

In South Carolina, any electrical work involving new wiring, panel modifications, or the addition of a transfer switch or interlock must be permitted under NEC Article 702 (Optional Standby Systems). This applies across Greenville, Pickens, and Spartanburg counties. The permit triggers a county inspection, which verifies:

- ✓ The interlock is properly installed and mechanically sound
- ✓ The wiring is correctly sized and properly terminated
- ✓ The inlet is correctly rated for weather and amperage
- ✓ The installation is fully backfeed-protected

What Happens If You Skip the Permit

- ✗ **Insurance exposure:** Your homeowner's insurance may deny a claim related to generator damage or an electrical fire if the installation wasn't permitted and inspected.
- ✗ **Home sale complications:** A permitted installation adds value and sails through a buyer's home inspection. An unpermitted one can be flagged and require remediation, at your expense, before closing.
- ✗ **Backfeed liability:** If a utility worker is injured because of backfeed from an unpermitted installation, the liability exposure for the homeowner is significant.

What the Inspection Process Actually Looks Like

Once the install is complete, the electrician files the inspection request with the county. A county inspector, not the electrician, visits the home, reviews the work, and either passes it or asks for a correction. A properly done installation passes on the first visit, typically within a few business days of the install. You don't have to be present, but you can be. The passed inspection paperwork is yours to keep, put it with your homeowner's insurance docs.

Permit fees in the three-county area typically run \$60-\$120. That's included in a complete quote. An electrician who asks you to pay permit fees separately, or who quotes you a suspiciously low number that doesn't account for permit costs, is either passing the cost to you after the sale or planning to skip it.

BPP HANDLES ALL OF IT

Backup Power Pro files the permit, schedules the inspection, and hands you a copy of the passed inspection report when the job is done. You never have to deal with the county building department. The install is compliant, documented, and covered, from day one.

5 Questions to Ask Before You Hire Anyone

Ask every electrician who quotes you. Any reputable contractor will answer yes to all five.

1

Are you a licensed SC Electrical Contractor?

Not just "licensed", specifically a licensed electrical contractor in South Carolina under the SC Contractors' Licensing Board. Panel work requires this license by state law. Ask for the license number and verify it at contractors.sc.gov. Handyman licenses, general contractor licenses, and out-of-state licenses don't qualify for panel work.

2

Does your quote include the permit and county inspection?

If it doesn't, ask why. In Greenville, Pickens, and Spartanburg counties, permits are required and run \$60-\$120. An electrician who excludes this is either adding it to the invoice later or planning to skip it entirely. Neither is acceptable.

3

Have you installed this in Greenville, Pickens, or Spartanburg county before?

Each county's building department has slightly different inspection processes and inspector preferences. Experience in the local market matters, especially for first-pass inspection success. Ask how many installs they've done in the area.

4

Will you give me an all-in price before I commit?

A reputable electrician can give you a firm, complete quote after reviewing your setup: panel location, generator specs, and the distance from the panel to the install point. "I'll give you the final number once I'm on-site" is a red flag. You should know the full price before anyone touches your panel.

5

What warranty do you offer on the installation?

Hardware is covered by the manufacturer warranty. But the installation itself, the wiring quality, the mechanical fit of the interlock, the weatherproofing of the inlet, should be warranted by the contractor. If they won't stand behind their work, that tells you something about the work.

BPP ANSWERS YES TO ALL FIVE

SC Contractor LIC #2942 • Permit and inspection included in every quote • Google reviews on our Business Profile • All-in price before you commit • Install passes inspection or we fix it free

YOU'VE DONE YOUR HOMEWORK.

Here's The Next Step.

Most homeowners who read this guide feel one of two things: relieved that this is simpler than they thought, or frustrated that they've had a generator sitting in the garage for years that could have been connected the right way. Either way, the next step is the same.

1

Start the Short Walk

A few questions and a panel photo at backuppowerpro.com/walk-v2. No technical knowledge required. Takes a few minutes from your phone.

2

Ballpark on the Walk, Firm Number After Review

You see a typical range for your amp choice right away. Key reviews your setup and texts the firm all-in number (inlet, interlock, cord, permit, inspection). No surprise add-ons after that text.

3

Done in One Day

A licensed SC electrician arrives, completes the connection, and the install passes county inspection before leaving your driveway.

[Get My Ballpark Price →](#)

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